

# An analysis of graph neural networks for fake review detection: A systematic literature review

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## ABSTRACT

Over the past decade, detecting fake reviews has emerged as a critical challenge in ensuring the credibility of online businesses. The capability of Graph Neural Networks (GNNs) to handle data structured in graphs is a key advantage, enabling a nuanced representation of the complex relationships among reviews, users, and products. With the growing demand for robust solutions, there has been a rising focus in the research community on integrating GNNs into fake review detection mechanisms. This research conducts a thorough systematic literature review covering GNN-based articles, conference papers, and scholarly works from 2019 to January 2024, examining the current state and challenges. Initially, we provide an introduction to the problem's background and a broad overview of spam review methods based on GNNs. Additionally, we highlight existing techniques for detecting fake reviews, specifically emphasizing GNNs-based models categorized by types. Subsequently, we compare key concepts, merits, and drawbacks of these methods within their respective categories. Finally, we outline several limitations and unresolved issues that merit future attention in this domain and present potential avenues for further research. We hope this study can serve as a valuable resource for researchers, system practitioners, and newcomers to overcome current challenges and navigate forthcoming scenarios when deploying a spam review detection system with GNNs.

## 1. Introduction

Advancements in internet technology have facilitated an increasing tendency to buy and sell products and services online. Since these products/ services are intangible for customers, their trust and credibility rely heavily on reviews before making purchase decisions. Businesses also depend on reviews to gauge customer satisfaction regarding their business, while manufacturers use reviews to understand customer preferences, product/service evaluations, and recommended improvements. Additionally, the significance of reviews is underscored by their impact on business performance, with positive reviews correlating with higher returns and vice versa [1–4]. According to Luca [5], increasing the star rating by at least one star can boost earnings by 5% to 9%. However, some businesses see this as an opportunity and hire spammers to generate fake reviews. A fake review is a dishonest or misleading evaluation of a product, service, business, or individual, typically generated with the intention to trick potential consumers or readers [6]. Fake reviews can originate from a variety of sources, such as company employees, business competitors, acquaintances, some member of family, or bots. Normally, the motives behind fake reviews include promoting

or demoting service/product reputation [7,8], personal vendettas, profits, etc. Dishonest reviews not only mislead consumers but also damage the reputation of products, brands, and the e-business as a whole. Fake reviews, categorized differently across various publications, typically fall into four common types, as summarized in Fig. 1.

The problem of fake reviews has been highlighted in various sources to demonstrate its severity. For instance: (1) Ellson [9] noted that one out of every three reviews on TripAdvisor is fake; (2) Alma [10] indicates that an estimated 11% to 15% of all reviews within three prominent product categories — consumer electronics, home and kitchen goods, and sports and outdoor equipment — are deemed to be fake; (3) Cavazos [11] reported that in May 2021, an open database disclosed the identities of over 200 000 individuals who appeared to be engaged in Amazon's fake product review schemes; (4) approximately 15% of reviews on e-commerce platforms fabricated [12]; (5) 8% to 15% of posted reviews on real-world websites are spam [13,14]. These alarming statistics have prompted researchers to investigate techniques for efficiently distinguishing between fake and genuine reviews.

The identification of deceptive reviews can be conducted through manual means, automated techniques involving traditional machine/

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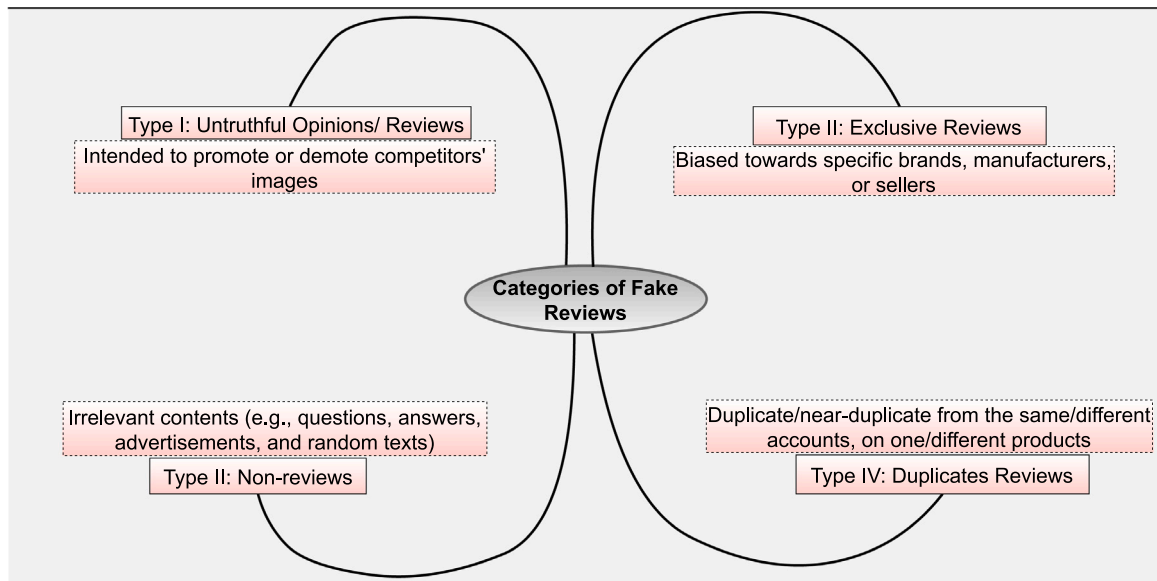


Fig. 1. Common categories of fake reviews.

deep learning (ML/DL), or a combination of both Salminen et al. [15]. Manual detection involves scrutinizing language structure variations, the use of brand names, and the presence of first-person pronouns (e.g., I, my). Studies indicate that fake reviews often demonstrate an increased frequency of first-person pronouns and brand names [16–18]. Over the past fifteen years, in an effort to eliminate the problem, researchers and experts have developed numerous automated approaches, ranging from traditional ML techniques to DL methods. These often involve analyzing review patterns, user behavior, and linguistic cues to identify suspicious content [19,20]. Although notable success has been achieved, the persistence of the problem is attributed to the economic benefits associated with fake reviews, leading spammers to strive for reduced detection likelihood. This sometimes involves crafting fabricated reviews to mimic genuine ones. Fig. 2 depicts the common fake review detection approaches.

Graph neural network based approaches have had remarkable success in research fields including natural language processing (NLP) [21], sentiment analysis [22,23], computer vision (CV), machine translation [24,25], object detection [26–28], and natural sciences [29], fake reviewer detection [1,30–32]. This success is primarily attributed to their ability to establish and represent complex interactive relationships among implicit and explicit features. In fake review detection context, graph learning has emerged as a prominent approach, driven by its efficacy in capturing intricate relationships within the richly interconnected data prevalent in such contexts. Firstly, the inherent complexity of fake review data, characterized by intricate relationships between users, items, attributes, and contextual factors, makes graphs an ideal representation [33–39]. The interconnected nature of these elements is especially apparent in online review systems, where various entities tightly associate and influence each other. Secondly, graph learning proves effective in unraveling complex relations and extracting knowledge from diverse graph structures [40–42]. This methodology aims to extract essential features from graphs, utilizing the acquired graph representations for specific tasks [43–47]. The diverse range of graphs generated from review system data significantly enhances the performance of fake review detection [4].

In the context of the disclosure matter, various graph-based literature surveys have been conducted, including GNNs for visual question answering (VQA) [48], deep learning-based graph anomaly detection [49], graph-based anomaly detection [50,51], graph-based methods for fake news detection [52], and graph learning [4]. However,

to the best of our understanding, no surveys on fake review detection have focused on GNN-based frameworks. Our survey specifically focuses on GNN-based methods, driven by the motivation to harness the aforementioned GNN benefits. Although certain similarities may be inevitable, our survey distinguishes itself from the mentioned works by concentrating on describing, analyzing, and discussing models for fake review detection employing the latest GNN-based techniques. Fig. 3 depicts a brief summary of how this study differs from existing similar works. In summary, the main contributions of this systematic literature are:

- We present an extensive literature survey on recent fake reviews techniques, encompassing related concepts, performance comparison insights based on different dataset characteristics, and datasets—associated collection techniques limitations. This survey serves as a practical guide, aiding in the clarification, enhancement, and proposition of diverse methods for detecting fake reviews.
- We offer a concise overview of the existing GNN models, including necessary comparisons among different types of models and summarizing their corresponding algorithms. To the authors' knowledge, this study represents one of the initial surveys concentrating on a GNN-based approach for the task of fake review detection
- We present and explore challenges pertaining to the detection and prevention of fake reviews through GNN models. A comprehensive analysis is conducted for each issue, and future research directions are proposed, with specific attention to the trade-offs between model depth and scalability.

The rest of this research is structured into six sections. Section 2 presents a brief background on graph neural networks (GNNs) and key related concepts relevant to this study. Section 3 outlines the systematic steps for executing the study effectively. In Section 4, we conduct a comprehensive survey of papers utilizing GNNs for fake review detection. It highlighted key aspects of the respective papers, including the addressed problems, employed methodologies, and the strengths and critiques inherent in each proposed solution. Section 5 discusses both general and specific observations covering limitations and potential future directions that need further attention. Finally, Section 6 provides a comprehensive summary of the key conclusions drawn from this study.

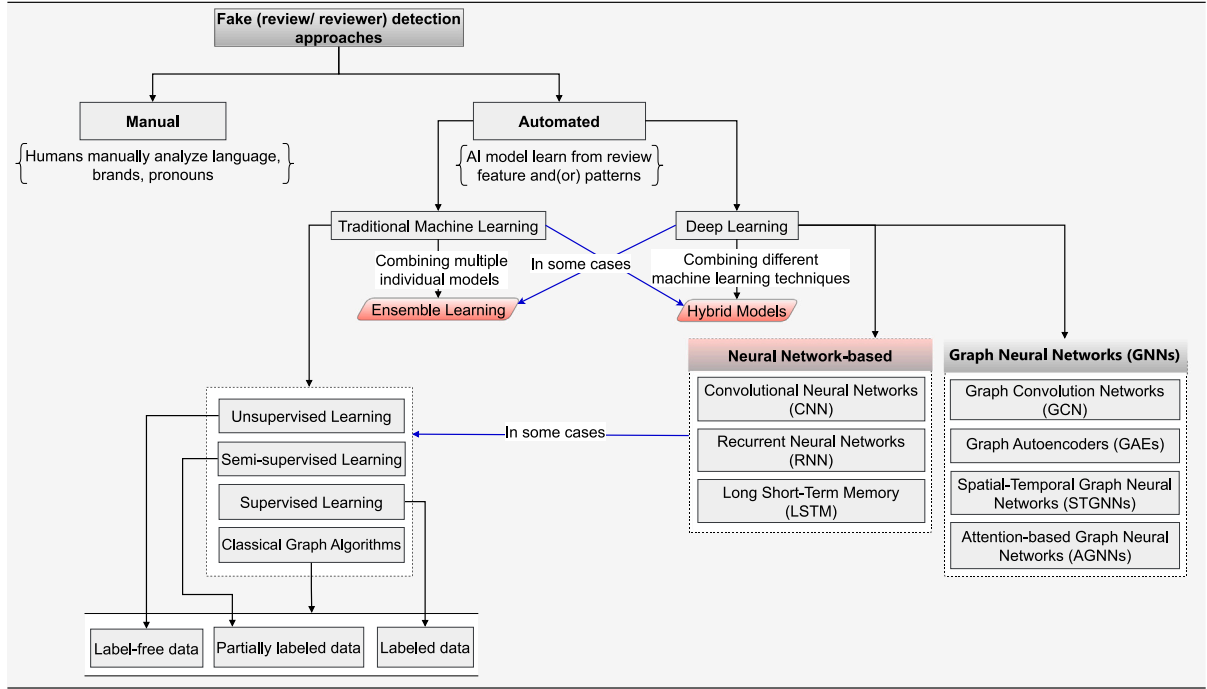


Fig. 2. Taxonomy of the common fake review detection approaches.

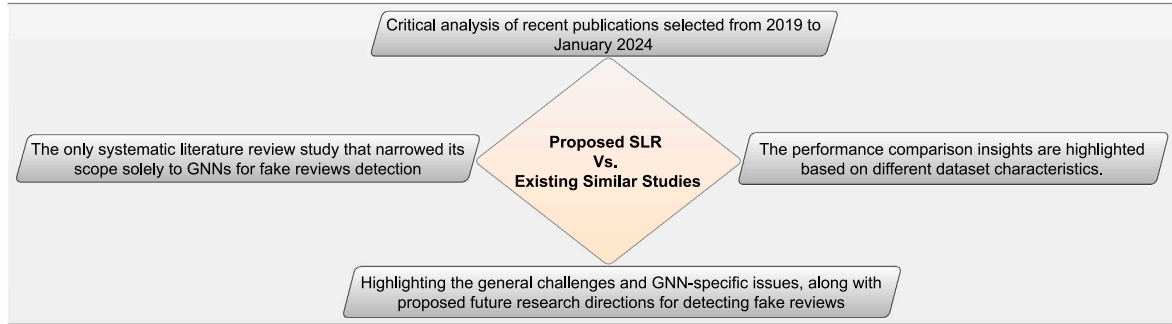


Fig. 3. The proposed systematic literature review versus existing similar works.

## 2. Background and preliminaries

In this section, we provide a brief background on graph neural networks (GNNs) and key related concepts.

### 2.1. Graph structure

**Conventional definition of graph structure:** Formally, structure of a simple graph is defined as  $G = (V, E)$ , and  $V = \{v_1, v_2, v_3, \dots, v_{n-1}, v_n\}$  is the set of nodes, and  $E = \{e_{1,1}, e_{1,2}, e_{1,3}, \dots, e_{1,n-1}, e_{n,n}\}$  is defined as the set of edges, where  $e_{i,j} = (v_i, v_j) \in E, 1 \leq i, j \leq n$ , where  $v_i$  and  $v_j$  are adjacent nodes. Then, an adjacent matrix  $M^a$  stands for  $n \times n$  expressed in Eq. (1).

$$M_{i,j}^a = \begin{cases} 1, & \text{if } e_{i,j} \in E, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Extending from simple graph structures, the literature proposes a notable number of improved categories [53–55]. The common categories include attributed graph structures (e.g., attributed and spatial-temporal graphs) and multi-relational graph structures (e.g., heterogeneous and multiplex graphs) [52].

- **Attributed Graphs:** An attributed graph  $G$  is a tuple  $G = (V, E, X)$  where  $V$  represents the set of nodes,  $E$  is the set of edges, and  $X$  is the attribute matrix. The attribute matrix  $X$  can be defined as  $X \in \mathbb{R}^{n \times d}$ , where  $n$  is the number of nodes, and  $d$  is the dimensionality of the attributes.
- **Spatial-Temporal Graphs:** A spatial-temporal graph  $G$  is represented as  $G = (V, E, T)$ , where  $V$  is the set of nodes,  $E$  is the set of edges, and  $T$  is the temporal component. The temporal component  $T$  is a function that associates each edge with a specific timestamp, capturing the temporal evolution of the graph.
- **Multi-relational Graphs:** A multi-relational graph  $G$  can be expressed as  $G = (V, E_1, E_2, \dots, E_k)$ , where  $V$  represent the set graph nodes, and  $E_1, E_2, \dots, E_k$  represent multiple edge sets, each corresponding to a unique relation type. The set of edges can be defined as  $E_i \subseteq V \times V$  for each  $i$ .
- **Heterogeneous Graphs:** A heterogeneous graph  $G$  is a complex structure denoted as  $G = (V_1, V_2, \dots, V_k, E_1, E_2, \dots, E_m)$ . Here,  $V_1, V_2, \dots, V_k$  are node types, and  $E_1, E_2, \dots, E_m$  are edge types. Each edge type  $E_i$  connects nodes of different types, reflecting diverse relationships in the graph.
- **Multiplex Graphs:** A multiplex graph  $G$  is characterized by  $G = (V, E_1, E_2, \dots, E_l)$ , where  $V$  is the set of nodes, and  $E_1, E_2, \dots, E_l$  are multiple edge sets or layers. Each edge set  $E_i$  captures a

distinct layer of interaction or relationship among nodes, allowing for a nuanced exploration of various modalities.

## 2.2. Graph neural networks

The concept of GNNs concept were first introduced in 2005, by Gori et al. [56], which was subsequently extended by Scarselli et al. [55] and Gallicchio and Micheli [57] in 2009 and 2010 respectively. Essentially, GNNs are constructed using DL models over graph structures to represent a category of neural network frameworks specifically to process and/or analyze graph-structured data; they serve as an extension of RNNs. The key components of a conventional GNN (CGNNs) include nodes, edges, neighborhood aggregation, and propagation rule. For example, let node  $v \in V$  is associated with a feature vector  $X_v$ , encapsulating its relevant attributes. The edges in  $E$  define connections between nodes, signifying relationships within the graph. GNNs operate through a series of layers, with a fundamental aspect being the aggregation of information from neighboring nodes. A common update rule for a node  $v_i$  at layer  $l$  involves aggregating features from neighboring nodes according to a specified propagation rule. Mathematically, this update can be expressed as:

$$h_i^{(l+1)} = f \left( \sum_{j \in \mathcal{N}(v_i)} \frac{1}{c_{i,j}} h_j^{(l)}, W^{(l)} \right) \quad (2)$$

where  $h_i^{(l)}$  represents the representation of node  $v_i$  at layer  $l$ ,  $\mathcal{N}(v_i)$  denotes the neighbors of  $v_i$ ,  $c_{i,j}$  is a normalization factor,  $W^{(l)}$  represents learnable parameters, and  $f$  is an activation function. This iterative process allows GNNs to capture complex patterns and dependencies within graph-structured data. In addition conventional GNNs can be classified into other taxonomies including: graph convolutional networks, graph autoencoders, and spatial-temporal graph neural networks, attention-based graph neural networks [53] as describe below:

- **Graph convolutional networks (GCNs):** GCNs were initially introduced in the work by Kipf and Welling [58]. After constructing the graph, the propagation function  $F$  is then defined as in Eq. (3), where  $\widehat{M}^a$  represents the normalized adjacency matrix, calculated

as  $\widehat{M}^a = C^{-\frac{1}{2}}(M^a)^T$ , where  $C$  is the diagonal degree matrix of  $(M^a)^T$ . The elements of  $C$  are computed as  $C_{ii} = \sum_j (M_{i,j}^a)^T$ . Additionally,  $(M^a)^T$  is derived from  $M^a$  by adding the identity matrix  $I$ , resulting in  $(M^a)^T = M^a + I$ . The iterative propagation process within a GCN at the  $l$ th convolutional layer unfolds as follows:

$$F(M^a, h^{(l)}) = f \left( \widehat{M}^a h^{(l-1)} W^{(l)} \right) \quad (3)$$

$$\begin{aligned} h^{(1)} &= f \left( \widehat{M}^a h^{(0)} W^{(1)} + \beta^{(1)} \right) \\ h^{(2)} &= f \left( \widehat{M}^a h^{(1)} W^{(2)} + \beta^{(2)} \right) \\ h^{(3)} &= f \left( \widehat{M}^a h^{(2)} W^{(3)} + \beta^{(3)} \right) \\ &\vdots \\ h^{(l)} &= f \left( \widehat{M}^a h^{(l-1)} W^{(l)} + \beta^{(l)} \right) \end{aligned} \quad (4)$$

where  $h^{(0)}$  is usually equal to  $X$ ,  $W^{(l)} \in \mathbb{R}^{m \times d}$ ,  $l = \{1, 2, \dots\}$ ,  $\beta$  is the bias.

- **Graph autoencoders (GAEs):** Graph autoencoders comprise two primary components: (1) the encoder, which transforms graph nodes into a latent feature vector space, and (2) the decoder, which reconstructs graph information from these latent feature vectors [59,60]. The function  $F$  is then redefined according to Eq. (3). However, in this context,  $M^a = f(O_e O_e^T)$  represents the reconstructed adjacency matrix of the decoder composition, where  $O_e$  corresponds to the result of the encoder composition.

The encoder step in GAEs, employs GCNs to generate the embedding matrix. Consequently, the calculation of  $O_e$  is based on Eq. (3), such that  $O_e = F(M^a, h^{(l)})$ , and  $O_e^T$  denotes the  $O_e$  transpose matrix.

- **Spatial-temporal graph neural networks (STGNNs):** STGNNs are designed to model inputs involving nodes with dynamic interdependencies. Many real-world tasks exhibit dynamism in graph inputs and structures. To accommodate this, an STGNN is introduced to appropriately represent such dynamic and interdependent data. STGNNs are usually categorized into two types: CNN- and RNN-based models.

**RNN-based:** In RNN-based model, spatial-temporal relations are captured by propagating the hidden states of STGNNs through a recurrent unit that utilizes graph convolutions [52,61]. The propagation function  $F$  of STGNNs is similarly illustrated as in Eq. (3). The calculation of the value for the  $l$ th layer is as follows:

$$h^{(l)} = f \left( W X^{n(t)} + \mathbb{E}^m h^{(l-1)} + \beta \right) \quad (5)$$

Here,  $X^{n(t)}$  denote the matrix node feature at time step  $t$ .

**CNN-based:** RNNs handle spatial-temporal graphs through recursive iterations, encountering limitations in terms of: propagation time, and gradient explosion/vanishing challenges [62]. In contrast, CNN-based approaches tackle these challenges through leveraging parallel computing, mitigating memory constraints, and ensuring the stable gradients.

- **Attention-based graph neural networks (AGNNs):** AGNNs eliminate all intermediate dense (fully connected) layers, replacing propagation layers with an attention that preserves the graph's structure [63]. Although AGNN incorporates graph attention layers, its propagation function is calculated in a similar way as expressed in Eq. (3). In every layer, they share a learnable linear transformation matrix  $L_T \in \mathbb{R}^{t_h \times d_h}$ ,  $t_h$  denote the number of the  $r$ th hidden layer, and  $d_h$  stands for the dimensionality of  $r$ th (hidden) layer, is applied to specific input feature of each node as expressed in Eq. (6); then row vector of node  $v_i$  can be expressed as in Eq. (7), where  $L_T^{(t-1)}_{(i,j)}$  is defined as in Eq. (8). The parameter  $\Lambda^{(t-1)} \in \mathbb{R}$  acts as a guidance mechanism for attention in propagation layers. It is important to note that the value of  $\Lambda^{(t-1)}$  in propagation layers varies with changes in hidden states.

$$h^{(t)} = f \left( L_T^{(t)} h^{(t-1)} \right) \quad (6)$$

$$h_{v_i}^{(t)} = \sum_{v_j \in \mathcal{N}(v_i) \cup \{i\}} L_T^{(t-1)}_{(i,j)} h_j^{(t-1)} \quad (7)$$

$$L_T^{(t-1)}_{(i,j)} = f([\Lambda^{(t-1)} \cos(h_i^{(t-1)}, h_j^{(t-1)})]) v_j \in \mathcal{N}(v_i) \cup \{i\} \quad (8)$$

## 3. Methodology

The research methodology section outlines the systematic steps followed in this study. It begins with defining the study scope and identifying the study questions, which involves setting study objectives and determining the specific RQs to be answered. The section then details the formulation of the search strategy to obtain relevant publications within the defined scope and the establishment of study selection procedures to guide the filtering of relevant publications addressing the study questions. Additionally, it covers the quality assessment of selected publications and the analysis of suitable information from the publications.

### 3.1. Research questions

This Systematic Literature Review (SLR) examines graph neural network (GNN) methods for detecting fake reviews from 2019 to January 2024. The study aims to thoroughly assess the effectiveness of these models in tackling deceptive review detection tasks. To achieve the study objectives, we formulated four research questions (RQs) to guide the process as follows:

- **RQ<sub>1</sub>** What are the existing graph neural network (GNN) approaches used for detecting fake reviews?
- **RQ<sub>2</sub>** Which publicly available datasets are commonly used to evaluate graph neural network (GNN) approaches for fake review detection? Do GNN models perform consistently across these datasets? If not, what are the reasons for the variations in performance?
- **RQ<sub>3</sub>** What general and specific challenges do graph neural network (GNN) frameworks face in effectively detecting fake reviews?
- **RQ<sub>4</sub>** What future research directions should be considered to effectively deploy graph neural networks (GNNs) in fake review detection?

### 3.2. Search strategy

In this study, we utilized two prominent indexing databases: Web of Science (WoS)<sup>1</sup> and Scopus,<sup>2</sup> as the main sources of the publications. These two databases are highly esteemed in the scientific community due standards and extensive coverage of publications indexed on other popular platforms, including the Institute of Electrical and Electronics Engineers (IEEE),<sup>3</sup> Springer,<sup>4</sup> the Association for Computing Machinery (ACM) Digital Library,<sup>5</sup> and ScienceDirect.<sup>6</sup> By integrating key search terms, we conducted a comprehensive exploitation on the existing GNN-based research in deceptive review filtering.

The main search terms included “fake review”, “spam review”, “deceptive review detection”, “review spam identification”, and “deceptive opinion spam”. These terms were chosen to comprehensively cover publications related to identifying spam reviews. Additionally, synonymous terms such as “review authenticity assessment” and “fraudulent review detection” were incorporated. The ‘OR’ operator was used to combine alternative expressions for each primary term, ensuring comprehensive coverage of key domain aspects, while the ‘AND’ operator concatenated these search strings. The search queries, along with specific limits applied in both databases, are as follows: (“graph neural network” OR “GNN” OR “graph convolutional network” OR “GCN” OR “graph-based model” OR “graph autoencoder” OR “GAE” OR “attention-based graph neural network” OR “AGNN” OR “spatial-temporal graph neural networks” OR “STGNN”) AND (“fake review” OR “spam review” OR “fake reviews detection” OR “review spam identification” OR “bogus review” OR “fraudulent review” OR “dishonest review” OR “deceptive review” OR “review spam” OR “manipulated review”) AND (LIMIT-TO(DOCTYPE, “conference papers”) OR LIMIT-TO(DOCTYPE, “reviews”) OR LIMIT-TO(DOCTYPE, “articles”)) AND (LIMIT-TO(SUBJAREA, “Engineering”) OR LIMIT-TO(SUBJAREA, “Mathematics”) OR LIMIT-TO(SUBJAREA, “Computer Science”)) AND (LIMIT-TO(LANGUAGE, “English”) AND PUBYEAR:[2019–2024]).

### 3.3. Study selection

The process of choosing qualifying publications serves to filter out those that are not explicitly focused on the detection of fake reviews using GNNs. To identify the pertinent papers, specific exclusion and inclusion criteria are established (as summarized in Table 1), including requirements such as being written in English, related to identification of deceptive reviews, including an experimental evaluation, having a publication from 2019 to January 2024, and having accessible full-text.

**Table 1**

Inclusion and Exclusion criteria.

| Inclusion criteria                                                                                                                                                                                                                                                 | Exclusion criteria                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Written in English</li> <li>• Related to spam review detection</li> <li>• Includes an extensive experimental evaluation</li> <li>• Studies published from 2019 to January 2024</li> <li>• Accessible full-text</li> </ul> | <ul style="list-style-type: none"> <li>• Exclusively presents mathematical models</li> <li>• Have no clear metadata such as publication information</li> <li>• Papers from workshops</li> <li>• Publications that include graph-based models but not related to fake reviews detection</li> <li>• Papers containing abstract only</li> </ul> |

Conversely, exclusion criteria involve those publications that exclusively present mathematical models, publications where publication information is not clearly indicated, papers workshops, articles that include graph-based networks but are not related to deceptive review detection, and papers containing abstracts only.

### 3.4. Synthesis of search results

Initially, the search queries resulted in the retrieval of 201 publications, which formed the basis for this focused review study. Adopting the evaluation process from our recent published work [64], we conducted a thorough assessment to include only suitable works. The evaluation procedures were deliberated during our regular weekly meetings among the authors and comprised three main steps:

- **Step 1:** Detecting and eliminating duplicate articles. This process aims to reduce redundancy and ensure the uniqueness and accuracy of the data used in the research.
- **Step 2:** Application of predefined (selection) criteria to the remaining publications. This ensured that only publications relevant to the RQs and which meet inclusion constraints were chosen for critical literature evaluation.
- **Step 3:** This step concentrates on evaluating the validity and quality check of the remained publications, ensuring that only related, valid, and suitable works related to the research scope are included for critical literature evaluation and analysis. A publication is labeled as *included* if and only if it meets the following criteria: it uses the GNN framework to address fake reviews, experimental design are well established, highlights key prevalent challenge in the field, and also provide the potential future work directions. Contrarily, works are considered *not suitable* if and only if it address fake reviews or closely related areas using other non-GNN approaches. Examples include general fake review detection, spam news classification, spam filtering, opinion spamming social networks, identification of spam user accounts, deception in discussion forums, and general aspects mining opinion, reputation management, or sentiment analysis. Publications that use methodologies not directly relevant to the core aspects of deceptive review identification, such as machine learning methods unsuitable for the task, are labeled as *out of scope*.

Following the quality assessment, 26 articles on GNNs for fake review detection were selected for detailed analysis. Fig. 4 illustrates the distribution of these publications by year, funding country, and GNN category. The analysis reveals the following insights:

1. The number of selected publications has generally increased, with peaks in 2020 and 2023 indicating growing research interest. The low number in 2024 is due to the collection of studies ended by January 2024 (i.e. it reflects the early stage of data collection for this year).

<sup>1</sup> <https://www.webofscience.com/>.

<sup>2</sup> <https://www.scopus.com>.

<sup>3</sup> <https://www.ieee.org/>.

<sup>4</sup> <https://link.springer.com/>.

<sup>5</sup> <https://www.acm.org/>.

<sup>6</sup> <https://www.sciencedirect.com/>.

**Table 2**  
GNNs-based clusters for fake review detection.

| Clusters         | Publications                                                                                                                                                                                                                                                             |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conventional GNN | Cheng et al. [65], Zhang et al. [66], Jayashree et al. [67], Andresini et al. [68], Liu et al. [69], Noekhah et al. [70], Liu et al. [71], Zhao et al. [32], Peng et al. [72], Chen et al. [73], Dou et al. [74], Liu et al. [75], Wang et al. [42] and Tang et al. [76] |
| GCN              | Li et al. [77], Wang and Wu [78], Li and Cheng [31], Xiang et al. [79], Kudo et al. [80], Wang et al. [81], Cao et al. [82], Zhang et al. [83], Zhu et al. [84], He et al. [85] and Huang et al. [86]                                                                    |
| AGNN             | –                                                                                                                                                                                                                                                                        |
| GAE              | –                                                                                                                                                                                                                                                                        |
| STGNN            | Shehnepoor et al. [87]                                                                                                                                                                                                                                                   |

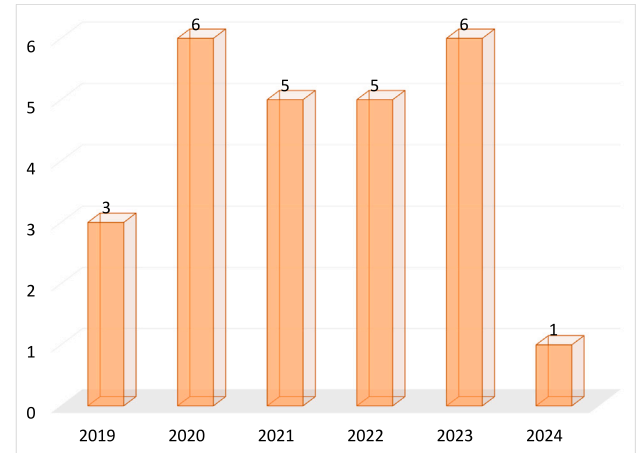
2. The data highlights CGNN and GCN as the primary focus of research within the provided GNN types, with significantly more publications compared to AGNN, GAE, and STGNN. This suggests a strong research foundation in CGNN and GCN approaches and a potential for further exploration in attention-based and autoencoder graph networks.
3. The geographic distribution highlights China as the leading contributor, followed by the USA, Japan, Australia, and other countries. This analysis emphasizes the global and collaborative nature of research in this field, revealing regional research priorities and opportunities for international cooperation.

#### 4. Critical analysis of selected studies

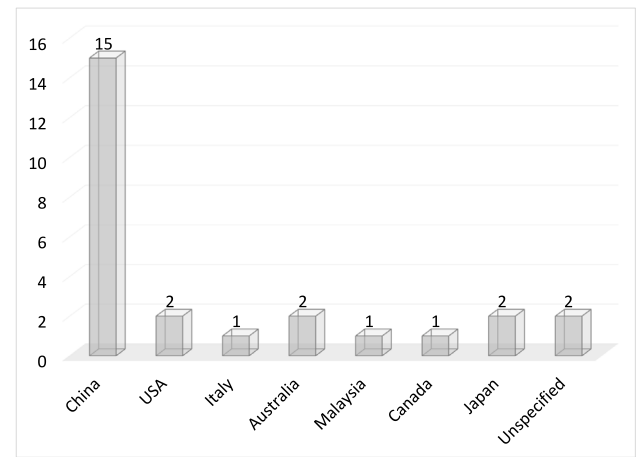
In this section, we present a survey of papers that utilize GNNs for fake review detection. We cluster them based on GNN taxonomies, as summarized in Fig. 2. Table 2 presents different categories (clusters) of existing GNN-based approaches for fake review detection, addressing  $RQ_1$  posed in Section 3.1. Within each cluster, we further describe the key aspects of the respective papers (as covered in subsections of Section 4), including the addressed problem, employed methodologies, and the strengths and limitations inherent in each proposed solution. Additionally, to answer the  $RQ_2$ : Do GNN models perform consistently across these datasets? If not, what are the reasons for the variations in performance? We added Table 3, which offers an overview of crucial elements such as the utilized datasets, performance metrics, possible reasons for higher or lower performances, and notable limitations identified. This structured exploration aims to provide readers with an insightful and comparative perspective on the diverse approaches adopted in GNN-based fake review detection studies

##### 4.1. Conventional GNN-based detection

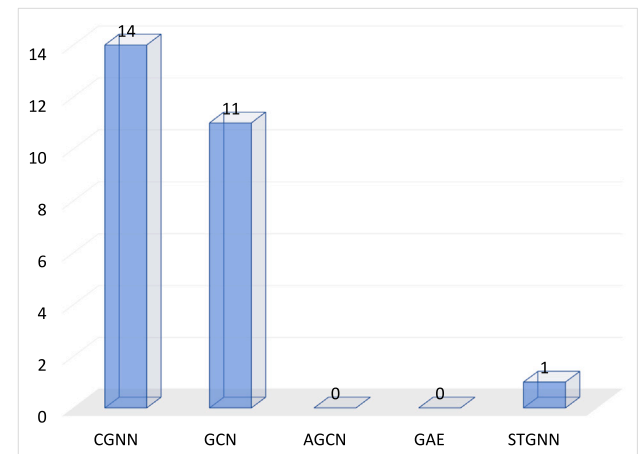
Cheng et al. [65] propose a GNN framework designed for the identification of fake reviewers in discussion forums. The methodology takes into account social interactions and author relationships, extracting information from diverse subgraphs formed by unique combinations of social networks. To address the imbalanced dataset issue, the social context network is crafted from user-post interactions through a repetitive training process. The framework encompasses a comprehensive social context graph, a homogeneous user-user subgraph, and a heterogeneous user-post subgraph. Emphasizing the importance of behavioral features over content in the realm of effective online fake review detection, the study underscores the critical necessity of incorporating social context and author behavior into the detection process.



(a) publication years



(b) funding countries



(c) GNN categories

Fig. 4. Publication distribution.

However, the paper lacks information on the generalizability of the proposed framework to different platforms or industries beyond the specific dataset used. Furthermore, it does not explore potential ethical implications or considerations related to the detection of fake reviewers and the use of social context information.

**Table 3**

Summary of selected conventional GNNs-based studies—Dataset, performance, and notable limitation.

| Paper                 | Technique | Datasets                                                          | Max. performances                                                                       | Limitations of the study                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|-----------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cheng et al. [65]     | GNN       | Collected from Taiwan's Fair Trade Commission (FTC) investigation | AUC = 0.904                                                                             | 1. Generalizability is constrained as it is only tested on a single-platform dataset<br>2. does not explore potential ethical implications                                                                                                                                                                                                                                                                      |
| Zhang et al. [66]     | GNN       | YelpCHI and Amazon                                                | Recall = 89.96% on Amazon and 74.58% on YelpCHI                                         | Concentrate exclusively on the detection of spam reviews, neglecting the pivotal aspect of identifying spam users                                                                                                                                                                                                                                                                                               |
| Jayashree et al. [67] | GNN       | Crawled from Yelp and Amazon                                      | Accuracy = 96%                                                                          | Not compared with more recent state-of-the-art models                                                                                                                                                                                                                                                                                                                                                           |
| Liu et al. [69]       | GNN       | YelpChi                                                           | AUC = 0.7428                                                                            | Result is low compared with recent similar state-of-the-art baselines.                                                                                                                                                                                                                                                                                                                                          |
| Noekhah et al. [70]   | GNN       | Synthetic crowdsourced dataset and Ott crowd sourced dataset      | Accuracy of 93% and 95.3% on the Synthetic crowdsourced and Ott's dataset, respectively | It only focuses on one or two types of spam entities and uses a limited number of features.                                                                                                                                                                                                                                                                                                                     |
| Andresini et al. [68] | GNN       | Hotel and Restaurant from Yelp                                    | AUC = 0.886 on Hotel, and 0.924 on Restaurant                                           | The approach focuses solely on individual features of users, products, and reviews, overlooking crucial mutual correlations between behavioral information, users comment, and ratings. Furthermore, it incorporates multiple interactions among users and service/product features, representing users-review-service/product and user-rating-service/product interactions through associated nodes and edges. |
| Liu et al. [71]       | GNN       | Collected from Dianping                                           | F1 = 0.81                                                                               | No clear analysis of node spamicity computation/node relations as it relate to their claims.                                                                                                                                                                                                                                                                                                                    |
| Zhao et al. [32]      | GNN       | Yelp and Amazon                                                   | AUC = 76.57% on Yelp, and AUC = 89.57% on Amazon                                        | No clear demonstration on how to they extract temporal features and explore fake reviewer communities.                                                                                                                                                                                                                                                                                                          |
| Peng et al. [72]      | GNN       | Yelp and Amazon                                                   | AUC = 83.54% on Yelp, and 94.44% on Amazon                                              | The assumption that spam reviewers usually comment on similar or the same products within a short time period and exhibit fixed pattern may not hold true                                                                                                                                                                                                                                                       |
| Chen et al. [73]      | GNN       | Crawled from the Mobile01.com                                     | AUC = 90%+                                                                              | Generalizability is questionable as it is only tested on one dataset                                                                                                                                                                                                                                                                                                                                            |
| Dou et al. [74]       | GNN       | Yelp and Amazon                                                   | AUC = 89.73% on Amazon, and 75.70% on Yelp                                              | No potential drawbacks or challenges associated with implementing the proposed label-aware similarity measure, reinforcement learning, or aggregation techniques provided.                                                                                                                                                                                                                                      |
| Liu et al. [75]       | GNN       | Yelp CHI, NYC, and Zip                                            | NDCG = 86.2% on CHI, 85.9% on NYC, and 89.6% on Zip                                     | Based on the assumption that subgroup membership can be accurately inferred using a GNN is not extensively discussed in terms of limitations and potential biases.                                                                                                                                                                                                                                              |

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In light of diverse deceptive stratagems and writing styles observed in spam reviews, two notable contributions have been made.

Firstly, Zhang et al. [66] has introduced a Metapath-based Subgraph for the purpose of detecting spam reviews. This innovative approach

Table 3 (continued).

|                   |     |                                                                                 |                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|-----|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wang et al. [42]  | GNN | YelpZip, YelpNYC, JD-Order, and JD-review                                       | AUC = 0.9783 on Zip, 0.9553 on NYC, 0.9712 on JD-Order, and 0.9573 on JD-review                                                         | 1. They exclusively emphasizes univariate characteristics of users, products, and reviews, neglecting vital interdependencies among behavioral signals, review textual content, and overarching ratings.<br>2. it integrates diverse interactions involving reviewers and product attributes, delineating reviewer-review-product and reviewer-rating-product relationships via associated nodes and edges within the model. |
| Tang et al. [76]  | GNN | Yelp and Amazon                                                                 | AUC = 92.56% On Amazon and 79.14% on Yelp                                                                                               | 1. the paper does not address the potential ethical and privacy concerns linked to spam detection on social media platforms<br>2. assume GNN construction commonly fails to filter out noisy neighbors without offering clear technical details, the paper lacks a comprehensive synthesis to support this claim.                                                                                                            |
| Li et al. [77]    | GCN | Crawled from Xianyu                                                             | AUC = 0.9872                                                                                                                            | Generalizability is constrained as it is based solely on data from a single platform.                                                                                                                                                                                                                                                                                                                                        |
| Wang and Wu [78]  | GCN | Tencent APP Store dataset                                                       | Precision = 0.95+                                                                                                                       | No provided details about the specific linguistic, behavioral, and graph patterns that are used to detect fake reviews and fraudsters.                                                                                                                                                                                                                                                                                       |
| Li and Cheng [31] | GCN | Hotel, Restaurant, and Doctor                                                   | Accuracy = 92.6% on Hotel, 94.6% on Doctor, 91.3% on Restaurant                                                                         | They exclusively rely on review text, disregarding key aspects of reviews and reviewer behavior.                                                                                                                                                                                                                                                                                                                             |
| Xiang et al. [79] | GCN | Yelp-hotel and Yelp-restaurant                                                  | Recall = 87.6% on Hotel, and 87.0% on Restaurant                                                                                        | They ignore: (1) important mutual correlations among behavioral information, users comment, and ratings, and (2) multi-interactions among the users and service/product features, namely as user-review-service/product and user-rating-service/product interactions.                                                                                                                                                        |
| Kudo et al. [80]  | GCN | Bitcoin OTC, Bitcoin Alpha, Epinions, Amazon App, Amazon Music, and Amazon Home | AUC = 0.996 on OTC, 0.97 on Alpha, 0.875 on Amazon, 0.973 on Epinions, 0.766 on Amazon App, 0.972 on Amazon Music, 0.756 on Amazon Home | Since the model's foundation is centered on the rating aspects of expanded balance theory, it is anticipated that the authors should provide a detailed analysis of expanded balance theory to enhance understanding of the rating's impact on performance.                                                                                                                                                                  |
| Wang et al. [81]  | GCN | Crawled from Tencent Inc.                                                       | F1 = 0.938                                                                                                                              | 1. generalizability is questionable as it focus solely on fraudster detection in app reviews, neglecting other online platforms<br>2. only mentions FdGars' ability to uncover malicious accounts without labeled data in anti-spam tasks without providing detailed insights.                                                                                                                                               |
| Cao et al. [82]   | GCN | YelpNYC, YelpZip, and YelpCHI                                                   | Precision = 0.7600 on NYC, 0.5641 on Zip, and 0.6667 on CHI                                                                             | Only mention of the need to reduce computational costs without providing specific strategies or optimizations to address this concern.                                                                                                                                                                                                                                                                                       |
| Zhang et al. [83] | GCN | Yelp and Movies & TV datasets                                                   | F1 = 0.992 on Yelp, and 0.985 on Movies & TV                                                                                            | Lacks a detailed discussion on the specific types of shilling attacks that the proposed framework, GraphRfi, is robust against.                                                                                                                                                                                                                                                                                              |

(continued on next page)

Table 3 (continued).

|                        |       |                                         |                                                                           |                                                                                                                                                                                                                                      |
|------------------------|-------|-----------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zhu et al. [84]        | GCN   | Hotel, Restaurant, YelpNYC, and YelpZip | AUC = 92.6% on Hotel, 92.1% on Restaurant, 70.3% on NYC, and 73.9% on Zip | The assumption that spammers consistently demonstrate uniform behavior, which may not always hold true.                                                                                                                              |
| He et al. [85]         | GCN   | YelpChi, YelpNYC, and YelpZip           | AUC = 87.69% on Chi, 87.85% on NYC, 88.05%                                | Ignores multi-interactions among users and service/product features, namely user-review-service/product and user-rating-service/product interactions, as it does not account for the mutual correlation between comment and ratings. |
| Huang et al. [86]      | GCN   | DBLP dataset from Jhy1993/HAN           | Accuracy = 0.8392                                                         | 1. Ignores mutual correlations between behavioral data, comment, and star ratings<br>2. lacks a comparative analysis with other existing fraud detection methods.                                                                    |
| Shehnepoor et al. [87] | STGNN | Yelp, and Amazon                        | Recall = 0.94 on Yelp, and 0.95 on Amazon                                 | The fixed window size for temporal information encoding is a limitation, as it may not capture the actual collusive behavior of reviewers involved in group fraud activities.                                                        |

NB: Overall, results show that models performance varies significantly due to three main reasons: (1) Imbalanced class distribution (the model achieved optimal performance on balanced datasets compared to imbalanced ones), (2) Dataset size (the larger the dataset, the lower the performance), and (3) Hybrid and multi-feature fusion-based models are more robust. Future work should consider addressing these issues.

integrates three metapath-based subgraphs — specifically, the user-item subgraph, user-review-item subgraph, and review subgraph — alongside syntactic information, aiming to enrich the pertinent representation of review information. Secondly, Jayashree et al. [67] and Andresini et al. [68] has undertaken a focused exploration into the extraction of implicit knowledge to automatically capture the intricate semantic meaning embedded in reviews within a heterogeneous network. Thirdly, Liu et al. [69] analyzed three types of inconsistencies (context, feature, and relation) and proposed solutions such as combining context embeddings with node features, filtering inconsistent neighbors, and learning relation attention weights. This involves a meticulous examination of top-n feature sets in a mutually reinforcing manner. However, certain limitations merit consideration in these studies. Firstly, the absence of comparative assessments with more recent state-of-the-art models employing pre-training techniques such as BERT and XLNET is noted. Secondly, both studies concentrate exclusively on the detection of spam reviews, neglecting the pivotal aspect of identifying spam users, which holds significance within the broader scope of the spam detection problem.

Noekhah et al. [70] proposes a graph-based method named ‘Multi-iterative Graph-based Opinion Spam Detection (MGSD)’ that considers all types of entities simultaneously within a unified structure to detect opinionated spam in e-commerce. Firstly, it initializes the entities and their spamicity scores and calculates the basic spamicity for entity samples. Then, it uses a multi-iterative algorithm to update entities’ spamicity scores and incorporates feature fusion techniques and machine learning algorithms to enhance detection accuracy. The algorithm applies a comprehensive set of domain-independent features and a new set of proposed features to improve detection accuracy. The reported results (i.e., an accuracy of 93% for the synthetic crowdsourced dataset and 95.3% for Ott’s crowdsourced dataset) are higher than its counterparts. However, (1) the proposed model relies on synthetic datasets for analysis and may not be applicable in real-world environments, (2) it only focuses on one or two types of spam entities and uses a limited number of features.

Liu et al. [71] presents an intricate probabilistic graph classification method designed for detecting opinion spam. To initially estimate the spamicity of nodes (including reviews, authors, and products) within the graph effectively, the authors train a neural network with an attention mechanism. This neural network learns a multimodal embedded

representation of nodes by utilizing both textual and rich features. Subsequently, leveraging the computed node priors, a heterogeneous graph is formed to capture relationships among different node types, and beliefs are iteratively updated through message propagation. The authors analyze various prominent rich features and the intermediate component, revealing that their states encapsulate specific statistical characteristics of the datasets. The reported results indicate superior performance compared to similar methods. However, there is no clear analysis of node spamicity computation/node relations as it relate to their claims.

In their works [32,72], frameworks based on heterogeneous GNNs are introduced to tackle the challenge of detecting stealthy fake reviewers by considering the interdependencies among reviewers, products/services, and reviews. The authors model fake reviewer detection using a heterogeneous graph comprising reviews, reviewers, and products to capture subtle fraudulent cues within their interactive relationships. They then introduce a Reinforced Neighbor Selector (RNS) to sample and aggregate top-k informative neighbors, effectively filtering out disguised relationships. Finally, a self-supervised heterogeneous graph embedding method based on mutual information is proposed to learn discriminative features for identifying fake reviewers. The presented results are promising, particularly in detecting well-disguised fake reviewers. However, the papers only mention that spam reviewers typically comment on similar or the same products within a short time period and exhibit fixed patterns. It is expected that future research will demonstrate how to effectively extract temporal features and explore fake reviewer communities.

To address the issue of astroturfing campaigns involving multiple collaborators, Chen et al. [73] proposes a detection framework based on GNN. This framework incorporates the stylometric patterns of the original perpetrators, including features such as review content, the combination of aspect category with review content, and title features. By examining the relationships among reviewers, the framework can identify the original posters of fraudulent reviews, revealing how they amplify each other’s perspectives to influence readers’ opinions. The effectiveness of the framework was evaluated against various DL models using fusion techniques, and the authors also discussed its managerial and theoretical implications. However, proposed model was only tested on one dataset.

Dou et al. [74] initially investigated the masking behavior of spammers and its adverse effects on GNN-based fraud detection systems. To

enhance these detectors against feature and relation mask by fraudsters, they introduced a label-aware similarity measure and a similarity-aware neighbor selector using reinforcement learning. In addition to two neural modules, they introduced a relation-aware aggregator to optimize computational efficiency. Experimental results highlight the impact of masked spammer on the performance of GNN-based fraud detectors. However, it is anticipated that the authors provide insights into potential drawbacks or challenges associated with implementing the proposed label-aware similarity measure, reinforcement learning, or aggregation techniques.

The study in [75] claims that GNN-based approaches suffer from the issue of discrepancy. To address the challenge of varying detection accuracy across different groups of reviewers on online review platforms, they propose a model aimed at recognizing and leveraging subgroup structures to enhance fairness in spam detection. The designed model can deduce concealed subgroup memberships, using them to adjust detection accuracy across groups and thereby improve fairness. Evaluation on Yelp review datasets illustrates the effectiveness of the proposed approach. However, the paper focuses on subgroup structures within reviewer groups and their influence on fairness in spam detection, omitting exploration of other possible factors or variables contributing to detection accuracy discrepancies. Furthermore, the model's reliance on the assumption that subgroup membership can be accurately inferred using a graph neural network is not extensively discussed in terms of limitations and potential biases.

Wang et al. [42] introduces a "Community-based Framework with Attention mechanism (C-FATH)" designed for large-scale heterogeneous graphs, integrating both heterogeneous and homogeneous graphs for fraud detection. The proposed algorithm incorporates community tags for candidate nodes, particularly focusing on eliminating the negative impact from non-similar neighbors within the same community with similar behavior patterns when constructing a node's neighborhood for heterogeneous graphs. The model's evaluation extends to two public review datasets and two large-scale datasets from JD.com. However, the effectiveness of the model remains uncertain, as there is no detailed comparison with other state-of-the-art approaches in terms of performance metrics and computational efficiency.

The research described in [76] presents an approach for identifying spammers on social media platforms utilizing graph neural networks (GNNs) and multi-relational graphs. To tackle the challenge of noisy neighbors in GNN construction, the method filters out deceptive and noisy nodes surrounding the central node. The model introduces techniques for intra-relation and inter-relation aggregation, aiming to improve the accuracy of spammer detection. The proposed model demonstrates superior performance compared to existing GNN-based spam detection methods when evaluated on Yelp and Amazon datasets. However, there are notable limitations: (1) the paper does not address the potential ethical and privacy concerns linked to spam detection on social media platforms, and (2) while asserting that GNN construction commonly fails to filter out noisy neighbors without offering clear technical details, the paper lacks a comprehensive synthesis to support this claim.

#### 4.2. GCN-based detection

Li et al. [77], the authors present comprehensive solutions to combat spam on a large scale, addressing two significant challenges: (1) handling the scalability of data and (2) countering adversarial actions by spammers targeting Xianyu, the largest second-hand goods app in China. The proposed approach, named the GCN-based Anti-Spam (GAS) model, utilizes Graph Convolutional Networks (GCN) to detect spam advertisements. This model integrates both heterogeneous and homogeneous graphs to capture both the local and global context of a comment. The algorithm operates on a bipartite graph with edge attributes specifically designed for Xianyu. However, the generalizability is constrained as it is based solely on data from a single platform.

In their work [78], a system for detecting, defending against, and forecasting adversarial fraudsters is proposed. This system aims to reveal camouflage accounts in an online app review platform. The detection process begins by selecting a small set of seeds with high precision using text and behavior features. Subsequently, a graph-based detection model is constructed to unveil hidden (distant) users who exhibit structural similarity to the seeds, employing the GCN algorithm. The reported evaluation precision, tested on the Tencent APP Store, exceeds 0.94. However, the paper does not provide details about the specific linguistic, behavioral, and graph patterns that are used to detect fake reviews and fraudsters.

The paper [31] proposes a detection model named SGCN-BERT, which combines the Semantic GGCN and pre-trained BERT framework to identify deceptive reviews. The model improves graph information by integrating semantic relationships and co-occurrence among words when constructing the heterogeneous graph. It demonstrates superior performance in comprehensive tests on four public datasets compared to previous methods. However, they depend solely on review text, overlooking important aspects of reviews and reviewer behavior. The drawbacks of relying excessively on text have been criticized in various studies. Ren and Ji [88] highlighted the challenge of extracting semantic meaning from reviews using linguistic features alone due to sparsity issues. Moreover, depending solely on review text features is ineffective, as studies, such as Budhi et al. [89], have shown no significant distinction between words in fake and genuine reviews, particularly on real-world commercial websites.

To overcome the cold-start challenge in deceptive review detection, Xiang et al. [79] proposes a feature fusion method. Initially, they use a GCN to combine behavior features, forming behavior association features (BAFs) by capturing mutual behavioral features information in the review. Then, a co-attention network integrates linguistic features and BAFs globally, addressing the lack of sensitive features in new reviews during cold-start scenarios. Leveraging a heterogeneous graph modeling user-product relationships, the method incorporates GCN to include mutual behavioral information. Additionally, BERT enhances the representation of review text at the sentence level. However, they overlook (1) correlations between behavioral feature information, customer comments, and star ratings, and (2) multi-interactions between users and service/product features, such as user-review-service and user-rating-service interactions. Additionally, Kudo et al. [80] suggest an end-to-end framework for fraudulent user detection using a GCN-based approach, incorporating an expanded balance theory that considers both the signs and directions of edges. However, as the model's foundation is centered on the rating aspects of expanded balance theory, it is anticipated that the authors should provide a detailed analysis of expanded balance theory to enhance understanding of the rating's impact on performance.

The paper by Wang et al. [81] introduces the FdGars framework, utilizing GCNs for detecting fraudsters in online app review systems. This involves combining text, behavior, and relationship features of reviewers by extracting content and behavior features and constructing a graph to capture fraudsters' relational dynamics with normal users. The study also explores detecting different types of fraudsters in online review systems, categorizing them based on content and behavior characteristics across various platforms, with results demonstrating effectiveness and scalability. However, some notable weaknesses include a focus solely on fraudster detection in app reviews, neglecting other online platforms. The evaluation of FdGars is limited to the Tencent App Store dataset, potentially impacting generalizability. The paper mentions FdGars' ability to uncover malicious accounts without labeled data in anti-spam tasks without providing detailed insights. Lastly, ethical implications and privacy concerns related to fraudster detection in online app reviews are not discussed.

Cao et al. [82] introduces an unsupervised and end-to-end method for identifying fake reviewer groups in online reviews. The approach

involves two main steps. Initially, modularity-based Graph Convolutional Networks (GCNs) are employed to detect cohesive groups. Subsequently, the suspiciousness of each group is assessed using anomaly indicators at both individual and group levels, leading to the identification of fake reviewer groups. Additionally, Zhang et al. [83] introduced a unified framework for both recommendation and fraudster detection, addressing the issue of fake feedback in recommender systems. The framework utilizes a GCN to learn user representations and incorporates the probability of a user being a fraudster in the recommendation component. Simultaneously, the prediction error in the recommendation component is utilized as a feature in the fraudster detection component. Experimental results demonstrate their superiority versus baselines. Limitations of the papers: (1) only mention of the need to reduce computational costs without providing specific strategies or optimizations to address this concern, (2) Zhang et al. [83] in particular, lacks a detailed discussion on the specific types of shilling attacks that the proposed framework, GraphRfi, is robust against.

Motivated by the expertise-driven construction of valuable features and the information aggregation mechanism of GCN, the authors in [84] introduce a graph-based approach for automatic feature extraction. This method capitalizes on the inherent structure of a review graph to generate features. The key premises of their approach include (1) the dependence of useful features on the dataset; (2) the potential influence of user behavior by their neighbors; and (3) the context-dependent nature of the property, such as whether a review is spam, relying on its surroundings. At the decision level, they propose an attention-based ensemble model to integrate the detection outcomes from various classifiers. The effectiveness of the proposed fusion model is validated through evaluations on multiple public datasets, showcasing promising experimental results. However, a limitation of this study is its assumption that spammers consistently display uniform behavior, which may not always hold true.

In their work [85], a spam review detection framework is introduced, which leverages graph convolutional networks (GCNs) in a heterogeneous graph context. This approach aims to capture diverse review-related metadata, including the review text, user features, and item features, as well as relational data represented in the form of a network. The GCNs are utilized to extract semantic information from the intricate interactions within this heterogeneous graph. The authors also incorporate a fusion technique to amalgamate low- and high-order interactions, combining propagated information across multiple subgraphs associated with reviews. Experimental results on publicly available datasets demonstrate the effectiveness of the proposed model, surpassing several strong state-of-the-art baselines. Nevertheless, a notable limitation of the method is the oversight of multi-interactions between users and service/product features, namely user-review-product/service and user-rating-service/product interactions, as it does not account for the mutual correlation among comment (in form of text) and overall (in form of star) ratings.

The study detailed in [86] delves into the application of a multi-layer GNN to identify fraudulent activities within rating systems, aiming to capture higher-order node information for more precise detection outcomes. The paper also addresses the challenge of gradient disappearance that can arise when employing a multi-layer GNN. The research explores the feasibility of employing varying numbers of hidden layer neurons and activation functions. Experimental evaluations on authentic datasets reveal that employing a multi-layer graph convolutional network leads to an enhancement in fraudster detection accuracy by up to 14.6%. However, the study has limitations in that it does not account for important mutual correlations among behavioral data, overall rating, and user comment. Additionally, the paper lacks a comparative analysis with other existing fraud detection methods, hindering a comprehensive assessment of its effectiveness in comparison to alternative approaches.

#### 4.3. STGNN-based detection

The conventional approach of depicting static networks for fraudster groups tends to neglect a reviewer's longitudinal behavior, rendering it ineffective in identifying outlier reviewers. To overcome this limitation, Shehnepoor et al. [87] proposes a spatio-temporal graph-based framework for detecting fraudster groups by considering temporal relations. The model leverages the efficacy of HIN-RNN in learning representations of reviewers and capturing their collaboration. Specifically, HIN-RNN models the coreview relations among reviewers within a group during a fixed time window. Subsequently, an RNN addresses spatial relations, predicting spatio-temporal relations among reviewers in the group. In the third step, a GCN refines vector representations of reviewers using these predicted relations. These refined representations are then utilized to identify and exclude outlier reviewers. The average representation of the remaining reviewers is fed into a simple fully connected layer to predict whether the group is a fraudster group. While the results are good, the fixed window size for temporal information encoding is a limitation, as it may not capture the actual collusive behavior of reviewers involved in group fraud activities.

#### 5. Limitations and future research attention

Despite the increasing popularity of GNNs in recent years, their adoption in the detection of fake reviews remains limited, notwithstanding promising results. To enhance the effectiveness of GNN-based fake review detection, it is crucial to consider the aforementioned issues, along with the following general and GNN-specific observations that require further attention from researchers. This subsection answers RQ<sub>3</sub> and RQ<sub>4</sub> posed in Section 3.1.

**General observation 1:** Based on this literature survey, it becomes evident that the most successful approaches involve hybrid and multi-feature fusion, encompassing both review- and user-centric aspect features, and incorporating various review characteristics related to content, context, and propagation. However, no study has explicitly explored the optimal or reasonable set of aspect or features suitable for integration, as the model's effectiveness may be slightly affected by the consideration or omission of certain review features [20,90]. Additionally, some previous studies view review burstiness as a reliable indicator of spam [8,91,92]. However, they fail to statistically analyze the optimal distribution rate, the ratio of reviewers' positive/negative sentiments (emotions) in reviews, and the rating distribution.

**General observation 2:** Over the last decade, social media has revolutionized the way businesses interact with consumers by offering a direct communication channel that enables them to engage with their audience on a personal level [93,94]. However, existing studies are still limited to detecting fake reviews on e-commerce websites or dedicated review platforms, often ignoring social media platforms. Reviews on social media usually contain multimedia information (images, text, video, or a combination), which significantly differs from reviews on dedicated business websites that typically contain simple features such as text, ratings, and emojis. Additionally, due to recent advancements in multimedia editing tools, detecting fake information has become more complicated as these tools provide easy methods for fabricating or modifying product or service content, such as image or video forgery [95]. In this regard, future work should focus on detecting fake business promotions on social media platforms.

**General observation 3:** Datasets are essential for the training and testing of any machine learning model. However,

- Many studies were evaluated using self-annotated datasets that were not publicly accessible. This may hinder fair comparisons with other research and benchmarking against existing models, as annotations may inadvertently introduce bias based on annotators' perspectives or expertise.

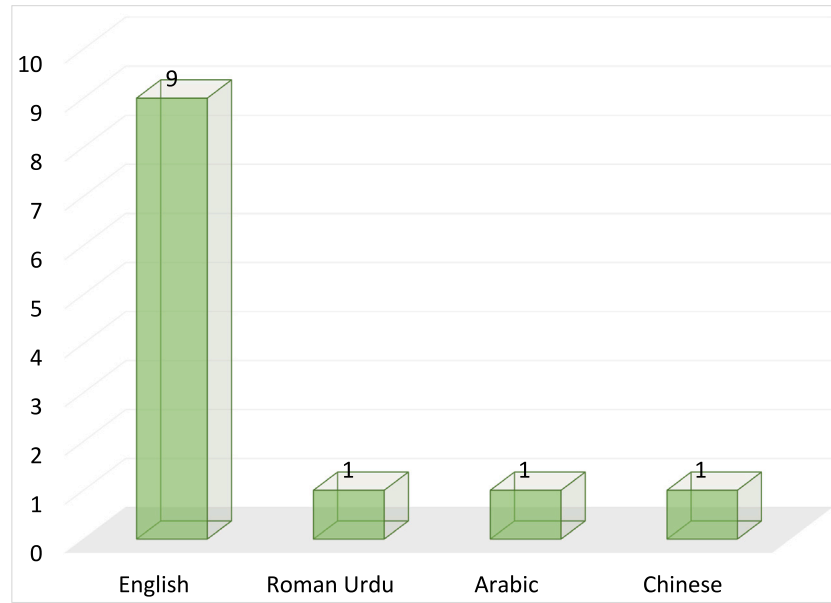


Fig. 5. Most popular public datasets distribution.

- The majority of benchmark datasets were gathered over the last two decades. They may not be appropriate given recent technological advancements, such as the emergence of large language models. For example, with tools like ChatGPT, spammers can easily write reviews that mimic genuine ones. Additionally, most existing gold-standard datasets often lack multidimensional review features beyond textual content, such as metadata required for GNNs and related frameworks [96].
- Most of the datasets were collected by adopting manual labeling through Turkers. Datasets collected through this approach are questionable, as Turkers may lack domain knowledge. Additionally, it is impractical to assume that all Turkers share the same psychological state of mind as real spammers [64]. In an attempt to address the issue of relying on questionable gold-standard datasets, researchers have turned their attention to semi-supervised or unsupervised techniques, as these require few or no labeled data, respectively. However, they still suffer from lower performance compared to supervised methods.
- There is a notable deficiency in multilingual and code-mixed review datasets. Code-mixed reviews are prevalent in countries with official dual languages, where reviews often incorporate a blend of languages [64]. Our research reveals a scarcity or complete absence of datasets and studies focused on fake review detection in several widely spoken non-English and low-resource languages, including Chinese, Spanish, Arabic, and Swahili. Fig. 5 illustrates the distribution of the most commonly utilized public datasets.

**GNNs specific observation 1:** Many conventional GNNs commonly employ undirected graphs with binary edge weights, which may not be suitable for various NLP tasks. For instance, in graph clustering, the objective is to find a partition that minimizes the difference in edge weights between dissimilar groups while maximizing the difference in weights among similar groups. If edge weights are binary, solving this problem becomes unfeasible. Future research endeavors should focus on constructing graphs where edge weights represent actual values, capturing the true relationships among nodes to a greater extent. Additionally, the influence of social network connections on opinionated spam activities, and the potential to detect spammers based on their connections and activities on social media, remain unexplored. Certainly, Cheng et al. [65] considered the problem with promising

performance. However, (1) generalizability is questionable since it is evaluated in a single platform or industries dataset, and (2) it does not explore potential ethical implications or considerations related to the detection of fake reviewers and the use of social context information.

**GNNs specific observation 2:** According to Phan et al. [52], GNNs are not well-suited for capturing the context of node features, including semantic relationships among phrases in an entire review text (sentence/paragraph). Therefore, we also concur that future research aimed at enhancing GNN-based models should prioritize the incorporation of node features derived from sentence embeddings or meaningful phrase embeddings. Additionally, the majority of GNN approaches exhibit a static structure, which poses challenges for real-time updates [53]. Hence, the construction of dynamic graphs capable of adapting spatiotemporally to real-time nature of reviews information becomes essential. In addressing this challenge [32] introduced temporal heterogeneous graphs and developed a dynamic community detection algorithm to capture temporal features and identify communities of fake reviewers. However, designing a low-latency reinforcement learning policy remains an open challenge. Furthermore, optimizing the learning policy in the reinforcement learning process is computationally expensive.

## 6. Conclusion

In this work, we extensively reviewed the literature on fake review detection. We narrow our focus to GNN-based studies on fake review detection, selecting studies from 2019 to January 2024. The introduction covers the central motivation behind the study, highlighting key contributions that distinguish it from similar studies. We then discuss the fundamental aspects of fake reviews, research methodology, study comparisons, and datasets summary, highlighting weaknesses in the data collection processes.

The number of selected publications has generally increased, peaking in 2020 and 2023, indicating growing research interest. Within GNN research, CGNN and GCN dominate, surpassing AGCN, GAE, and STGNN, suggesting a solid foundation and potential for further exploration in attention-based and autoencoder graph network frameworks. Moreover, we highlighted the global and collaborative nature of the field, with analysis results showing that China leads in research contributions, followed by the USA, Japan, Australia, and others. Additionally, we noted the lack of datasets and studies on fake review detection in non-English and low-resource languages. Regarding

the model's performance, notable variability is attributed to imbalanced data distribution and size. Furthermore, hybrid and multi-feature fusion-based methods have achieved state-of-the-art performance.

We also present the theoretical and mathematical foundations to ensure the necessary background and principles behind GNNs are well-covered to support the readers' understanding. Finally, we provide both general and specific observations covering limitations and potential future directions that need further attention. Thus, we hope this study will provide rich information and encourage researchers interested in further exploring the field, particularly regarding the application of GNNs in detecting fake reviews.

### CRedit authorship contribution statement

**Ramadhani A. Duma:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhendong Niu:** Supervision, Project administration, Funding acquisition, Conceptualization. **Ally S. Nyamawe:** Writing – review & editing, Methodology. **Ali Asghar Manjotho:** Writing – review & editing, Software, Methodology, Formal analysis.

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### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

No data was used for the research described in the article.

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